

Zijue Chen

Melbourne, Australia | zijuechen.36@gmail.com

Passionate researcher and educator working at the intersection of AI, robotics, automation, and smart manufacturing, with a research focus on adaptive AI for real-world engineering systems. My work investigates how AI-enabled systems can efficiently acquire, transfer, and adapt engineering knowledge as operating conditions change, and translate this knowledge into reliable monitoring, decision-making, and control of physical processes. This research spans data-efficient and few-shot learning, intelligent sensing, digital twins, industrial communication and closed-loop deployment, with applications across advanced manufacturing and robotic systems. I am particularly interested in developing AI that evolves with engineering systems rather than remaining a static, offline predictive tool.

AREAS OF EXPERTISE

AI, Robotics and Intelligent Systems

- Machine learning, computer vision, robotics and automation, intelligent sensing, monitoring and control, data-efficient and few-shot learning

Smart Manufacturing and Digital Twins

- Digital twins, Industry 4.0 systems integration, OPC UA and industrial communication, manufacturing data infrastructure, process monitoring and optimisation

Additive Manufacturing and Engineering Deployment

- Directed energy deposition (DED), laser powder bed fusion (LPBF), industrial robot programming, real-world AI deployment, industry-engaged engineering research

EDUCATION

Monash University **Melbourne, Australia**
Doctor of Philosophy: Robotics and Mechatronics 2018–2022

- **Thesis:** Development of a Deep Learning-Based Machine Vision System for Occluded 2D Fruit Trees.
- **Awards:** Monash Graduate Scholarship, Monash International Tuition Scholarship, and RTP Stipend Scholarship.

Monash University **Melbourne, Australia**
Bachelor of Engineering in Mechanical Engineering (Hons) 2016–2018

- **WAM:** 86.273%; **GPA:** 3.75/4.00.
- **Award:** The Faculty of Engineering Dean's Honours List (2016, 2017).

Central South University **Changsha, China**
Bachelor of Engineering in Mechanical Engineering 2013–2015

- **WAM:** 90.02%.
- **Awards:** First Prize Scholarship, Corporate Scholarship, and Outstanding Student Award (2014); Third Prize Scholarship (2015).

PROFESSIONAL EXPERIENCE

CSIRO **Melbourne, Australia**
CERC Postdoctoral Research Fellow 2024–Present

- Lead research on data-efficient and deployable AI methods for additive manufacturing, with a focus on process-condition drift adaptation, melt-pool representation learning, and real-time decision support for future closed-loop control.
- Lead an internal CSIRO collaboration on OPC UA-based communication for real-time defect detection and control, integrating heterogeneous manufacturing subsystems across sensing, robotics, and process equipment.
- Co-supervise a Monash final-year student team developing retrieval-augmented large language model agents for manufacturing systems, enabling machine-specific knowledge access, parameter guidance, troubleshooting, and operator training for DED and multi-material LPBF platforms.
- Contributed to a strategic leadership proposal during CSIRO's internal reorganisation, with primary input into digital-twin and role-division components for integrating Data61 and Manufacturing capabilities.

Monash University **Melbourne, Australia**
Research Fellow 2021–2023

- Supervised more than 30 final-year, Master's, and PhD students across machine vision, additive manufacturing, and smart manufacturing projects.
- Served as a founding contributor to the Monash Smart Manufacturing Hub, leading substantial elements of its AI, digital-twin, data, and communication capability development from the ground up.
- Conducted research on deep learning, computer vision, and digital-twin applications for additive manufacturing, including in-situ monitoring, defect detection, warping prediction, and process optimisation.
- Contributed to industry-engaged research and project development with partners including KUKA, Hexagon, and InfraBuild, supporting the translation of smart manufacturing and automation research into applied engineering contexts.

Monash University**Melbourne, Australia****Teaching Assistant****2018–2023**

- Delivered undergraduate and postgraduate teaching across Robotics (TRC4800), Systems Engineering (TRC3000), Lean Manufacturing (MEC5897), and Automation Design Project (TRC4407) through laboratories, tutorials, workshops, project-based learning, and direct student consultation.
- Contributed extensively to curriculum and teaching-material development, including the design of laboratories, workshops, tutorials, assignments, milestone assessments, and examination materials; played a substantial role in developing the teaching materials and project structure for the Automation Design Project unit.
- Achieved strong student feedback in teaching activities, including *SETU scores of 4.5/5 for TRC4800 Robotics and 4.74/5 for MEC5897 Lean Manufacturing*.
- Supervised and mentored undergraduate and postgraduate students through final-year research projects, practical engineering activities, and project-based learning, and coordinated the Monash Automation student team in the delivery of multidisciplinary engineering projects.
- Adapted laboratory and project-based teaching for online delivery during the COVID-19 pandemic, maintaining student engagement and support through Moodle, online forums, email, and individual consultation.

ARC Nanocomm Hub, Monash University**Melbourne, Australia****Research Assistant****2018–2019**

- Supported the experimental design, field deployment, and validation of robotics research projects, including repeated farm-based data collection and on-site testing in Australia and China.
- Conducted risk assessment and occupational health and safety (OHS) planning for laboratory and field-based research activities involving robotic systems, sensors, and outdoor trials.
- Represented Monash in technical demonstrations and external engagement activities related to robotics and agricultural automation projects, contributing to project visibility across industry, research, and public-facing forums.

SELECTED PROJECTS**AI-Based Control Strategies for Directed Energy Deposition (DED), CSIRO****2024–Present**

- Developed a highly data-efficient few-shot learning framework for rapid response-surface reconstruction under unseen process-condition drift in DED, enabling accurate prediction from as few as 1–3 observations at deployment and fewer than five observations per condition during training.
- Designed a domain-structured, latent-space-based adaptation pipeline that combines prior process knowledge with sparse new-condition observations, creating a practical basis for real-time parameter selection and future closed-loop process control.
- Developed self-supervised and domain-knowledge-guided learning pipelines for melt-pool feature extraction under limited-data conditions, producing compact and information-rich representations for downstream monitoring and decision-making tasks.
- Generated three first-author publications from this research, including a paper in *Additive Manufacturing*, an award-winning IEEE CASE 2025 paper on data-efficient representation learning for manufacturing monitoring, and another IEEE CASE 2026 paper on data-efficient condition-aware knowledge adaptation for geometry prediction.
- Extended this work toward practical deployment by leading the integration of closed-loop communication across the DED system stack (including robot, laser, and powder-feeder subsystems with mixed control protocols), and by supervising the development of a retrieval-augmented LLM tool to support machine operation, parameter selection, and troubleshooting using manuals and accumulated domain knowledge.

Industrial Oven Control Integration for Composite Curing, CSIRO / Boeing**2026–Present**

- Led the industrial control-integration work required to connect surrogate-model-based decision making with a live composite-curing oven, integrating Ignition HMI, OPC communication, Omron PLC recipe control, and onsite validation under real operating conditions.
- Established and validated the oven's real-time recipe-control pathway, demonstrating that active setpoints, ramp rates, and hold times can be modified during operation and propagated directly to PLC-controlled thermal processes.
- Identified and characterised key control-system constraints affecting closed-loop deployment, including PID behaviour, feedback-variable selection, actuator limitations, and PLC control logic, providing the technical basis for the next phase of safe autonomous process control.

Smart Manufacturing Integration and Digital Twin Development, Monash Smart Manufacturing Hub (MSMH)**2022–2023**

- Served as one of three founding contributors to the Monash Smart Manufacturing Hub, leading substantial elements of its additive manufacturing and digital-twin capability development from the ground up through the integration of AI modules, sensing systems, communication infrastructure, and manufacturing data workflows.
- Developed a vision-based deep learning system for 3D printing that used first-layer observations to predict future warping before failure became visible, enabling earlier intervention and reducing the cost of unsuccessful prints.
- Led multiple AI-enabled smart manufacturing functions, including process monitoring, defect detection, failure prediction, parameter optimisation, MES/SQL-based manufacturing data management, and OPC UA-based communication-layer integration for Industry 4.0-enabled collaborative tasks such as platform cleaning, part removal, and automated workflow coordination.
- Led the development of the service-oriented digital twin framework published in the *Journal of Manufacturing Systems*, and developed a strong practical view of digital twins as systems that should grow deliberately from minimal viable bi-directional information flow, with functionality added incrementally according to required decisions, enabling predictions, available data, and real process constraints.

Occlusion-Aware Machine Vision for Apple Harvesting Robots, Monash University 2018–2022

- Developed a highly use-case-specific machine vision pipeline for apple harvesting robots to infer continuous branch structures under natural occlusion, enabling distinction between hard occlusions (e.g. large branches) and soft occlusions (e.g. leaves and small twigs) for safer robotic path planning and fruit harvesting.
- Led the development of a novel neural-network-based strategy that used winter images of the same trees as structural references to improve inference of occluded summer branch geometry, substantially outperforming direct summer-only prediction and addressing a niche but critical vision challenge not well captured by conventional computer vision methods.
- Produced four Q1 journal papers from this research, including three first-author publications in *Computers and Electronics in Agriculture*, and contributed to additional work on reducing semantic-labelling effort through iterative training and optimisation strategies.
- Validated the vision system through repeated field data collection and on-farm trials in both Australia and China, including deployment on edge computing hardware for real robotic harvesting tests; the broader apple-harvesting robot project later received the **Engineers Australia 2023 National Winner of Excellence Award – Project of the Year**.

Patient-Specific 3D-Printed Prosthetic Hand Design, Monash University 2017–2018

- Led the mechanical design of low-cost, patient-specific 3D-printed prosthetic hands for two real users with partial finger loss, using hand scanning and custom design to create functional devices tailored to individual anatomy and daily-use needs.
- Designed a compliant TPU-based finger architecture with tendon-driven routing and joint-level geometric tuning to better approximate natural grasping trajectories, replacing more rigid gear-based designs with a lighter, cheaper, and more adaptive solution.
- Developed a coordination mechanism that allowed four prosthetic fingers to close to different extents while naturally balancing grasp force across digits, and introduced a hybrid TPU–silicone finger structure to mimic the roles of skeletal support and soft tissue deformation, substantially improving grip adaptability and friction.
- Designed and conducted user-centred functional evaluation tasks based on hand-therapy assessment principles, generating practical insights into how different users adapted to the prosthesis; this work contributed to two publications and received the **Amy Geach Hand Therapy Innovation Award**.

LPBF Porosity Prediction and Process Optimisation – AWS Cloud Computing Seed Project, Monash University 2022–2023

- Awarded an AWS Cloud Computing Seed Project to develop a conditional deep generative model linking LPBF process parameters to porosity, with the resulting surrogate model applied to process-parameter optimisation.

Sustainable Mobility and Cycling Safety – Victorian Government-Funded Project, Monash University 2023

- Developed a vision- and transformer-based framework for detecting cyclist near-miss events and translating camera observations into bird's-eye-view representations for safety and environmental analysis.

SELECTED PUBLICATIONS

- **Chen, Z.**, Wang, H., & Gunasegaram, D. (2025). MelpoolEmbedder: An information-rich, orientation-invariant deep learning framework for efficient melt pool feature extraction in metal additive manufacturing. *Additive Manufacturing*, 104900.
- **Chen, Z.**, Wang, H., & Gunasegaram, D. (2026). From few-shot to zero-shot: Condition-aware amortized response surface adaptation for data-efficient geometry prediction in metal additive manufacturing. *2026 IEEE 22nd International Conference on Automation Science and Engineering (CASE)*. Accepted for publication; presented at the conference.
- **Chen, Z.**, Wang, H., & Gunasegaram, D. (2025, August). Tackling the ‘small data’ challenge: A Versatile Melt Pool Encoder via Reconstructive Self-supervised Pretraining for Few-Shot Learning. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 729-734). IEEE.
- **Chen, Z.**, Surendraarcharyagie, K., Granland, K., Chen, C., Xu, X., Xiong, Y., ... & Tang, Y. (2024). Service oriented digital twin for additive manufacturing process. *Journal of Manufacturing Systems*, 74, 762-776.
- **Chen, Z.**, Granland, K., Tang, Y., & Chen, C. (2024). HOB-CNNv2: Deep learning based detection of extremely occluded tree branches and reference to the dominant tree image. *Computers and Electronics in Agriculture*, 218, 108727.
- Zhu, Y., **Chen, Z.**, Zhu, Y., Gresil, M., & Tang, Y. (2025). Bio-inspired perturbed hierarchical mechanical metamaterial for energy absorption. *International Journal of Mechanical Sciences*, 110847.
- **Chen, Z.**, Zhang, Z., Granland, K., Chen, C., & Tang, Y. (2023). Exploring Optimal Adaptive Process Parameters for Curved Infill Paths of Additive Manufactured Carbon Fibre Reinforced Polymers. *Composites Communications*, 39, 101549.
- **Chen, Z.**, Santhakumar, P., Granland, K., Troeung, C., Chen, C., & Tang, Y. (2023, August). Predicting Future Warping from the First Layer: A vision-based deep learning method for 3D Printing Monitoring. In *2023 IEEE 19th international conference on automation science and engineering (CASE)* (pp. 1-6). IEEE.
- **Chen, Z.**, Granland, K., Newbury, R., & Chen, C. (2022). HOB-CNN: Hallucination of Occluded Branches with a Convolutional Neural Network for 2D Fruit Trees. *Smart Agricultural Technology*, 100096.
- **Chen, Z.**, Ting, D., Newbury, R., & Chen, C. (2021). Semantic Segmentation for Partially Occluded Apple Trees based on Deep Learning. *Computers and Electronics in Agriculture*, 181, 105952.
- Granland, K., Newbury, R., **Chen, Z.**, Ting, D., & Chen, C. (2022). Detecting Occluded Y-shaped Fruit Tree Segments using Automated Iterative Training with Minimal Labelling Effort. *Computers and Electronics in Agriculture*, 194, 106747.

AWARDS, INVITED TALKS AND PROFESSIONAL SERVICE

- **Winner of Women in AI APAC Awards**, AI in Manufacturing Category, 2026
- **Peter Luh Memorial Best Paper Award**, IEEE CASE, 2025
- **CSIRO Research Excellence Award**, 2025
- **Invited Symposium Organiser**, *AI-Assisted Materials and Process Design*, PACRIM 17, 2027
- **Keynote Speaker**, *A Knowledge-Driven AI Assistant for Digital Twin Enablement in Metal Additive Manufacturing*, ACAM12, 2026
- **Invited Speaker**, *Novel Knowledge Adaptation Algorithm Enabling Rapid Change of Process Parameters in Metal Additive Manufacturing to Maintain Quality Requirements*, PRICM 2026
- **Invited Research Seminar**, Oak Ridge National Laboratory, 2025
- **Invited Seminar**, Monash Centre for Additive Manufacturing, 2025
- **Invited Seminar**, *Digital Twins for Advanced Manufacturing*, Deakin University, 2024
- **Invited Panelist**, AMTIL “AI in Manufacturing”, 2024
- **Invited Panelist**, Enjoy AI Education (Australia and global panel), 2024-2025
- **Co-Chair, CSIRO National CERC Committee**, 2025-2026; strengthened engagement with CSIRO leadership, co-led the establishment of state-based committees and Terms of Reference drafting.
- **Member, MF Staff Engagement and Culture Committee**, 2026
- **Organising Committee Member**, *Real-Time Decisions Using ML* Symposium, CSIRO, 2025
- Presented technical tours and seminars for CSIRO industry and government partners including **Boeing, Austal, and DSTG**
- **Industry Mentor, CSIRO Innovate to Grow mentoring program**, 2024–2026; mentored multiple local companies over eight-week engagements, helping translate practical manufacturing and automation challenges into actionable technical directions.
- **Google Cloud Research Credits Award**, 2023
- **AWS Cloud Seed Project Award**, Monash University, 2022

SKILLS AND INTERESTS

Technical Skills: Python, PostgreSQL, MATLAB, C, Java; machine learning, computer vision, OpenCV; SolidWorks; industrial robotics (KUKA, Yaskawa Motoman, UR); additive manufacturing (FDM, DED, LPBF).

Languages: English (Fluent), Chinese (Native).